Code No:R2322121 **R23** SET - 1

II B. Tech II Semester Regular Examinations, April/May– 2025

OPTIMIZATION TECHNIQUES

(Common to AI&ML, CSE(AI), CSE(AI&DS), CSE(AI&ML), CSE(DS), IT)

Time: 3 hours Max. Marks: 70

Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B)**
2. Answer **ALL** the question in **Part-A**
3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
4.All Questions carry **Equal** Marks

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PART –A (10x 2Marks=20M)

1.	a)	What are objective function contours?	(2M)
	b)	Write the Taylor's series expansion of a function f(x)	(2M)
	c)	Why is linear programming important in several types of industries?	(2M)
	d)	Explain the significance of Slack variable with example.	(2M)
	e)	How will you define transportation problem?	(2M)
	f)	How do you overcome redundancy in transportation problem.	(2M)
	g)	What are the limitations of classical methods in solving a one-dimensional minimization problem?	(2M)
	h)	Why is the steepest descent direction not efficient in practice, althoughthe directions used are the best directions?	(2M)
	i)	State the principle of optimality	(2M)
	j)	What sort of problems can be solved using dynamic programming?	(2M)

PART –B (5x10Marks=50M)

UNIT-I

2.	a)	i)What is graphical optimization, and what are its limitations? ii)How do you solve a maximization problem as a minimization problem?	(10M)
		OR	
	b)	i)State the Kuhn–Tucker conditions ii) Find the dimensions of a cylindrical tin (with top and bottom) made up of sheetmetal to maximize its volume such that the total surface area is equal to $A_0 = 24\pi$.	(10M)
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UNIT-II

3. a) Reduce the system of equations into a canonical system with x_1, x_2 and x_3 as basic variables. From this derive all other canonical forms. (10M)

$$2x_1 + 3x_2 - 2x_3 - 7x_4 = 2$$
$$x_1 + x_2 - x_3 + 3x_4 = 12$$
$$x_1 - x_2 + x_3 + 5x_4 = 8$$

OR

b) A television company has three major departments for manufacture of its two models A and B. Monthly capacities are given as follows: (10M)

Departments	Per unit time requirements(hours)		Hours available in this month
	Model A	Model B	
I	4.0	2.0	1600
II	2.5	1.0	1020
III	4.5	1.5	1600

The marginal profit of model A is Rs.400 each and that of model B is Rs.100 each. Assuming that the company can sell any quantity of either product due to favourable market conditions, determine the optimum output for the models, thehighest possible profit and the slack time in the three departments? Formulate asLPP model and solve it by simplex method.

UNIT-III

4. a) i)State the assumptions involved in the transportation problem. (10M)
ii)What are shadow prices in transportation problem? Explain.

OR

b) A company has factories at F1, F2 and F3 that supply products to ware houses at W1, W2 and W3. The weekly capacities of the factories are 200,160 and 90 units. The weekly warehouse requirements are 180,120 and 150/units respectively. The unit shipping costs in rupees are as follows. Find the optimal solution. (10M)

	W_1	W_2	W_3	Supply
F1	16	20	12	200
F2	14	8	18	160
F3	26	24	16	90
Demand	180	120	150	450

UNIT-IV

5. a) Min $f(x) = x_1^2 - x_1x_2 + 3x_2^2$. Starting point (1,2) by using steepest descent method.Show calculations for two cycles. (10M)

OR

b) Using penalty function method, solve the following problem. (10M)
Min $Z = (x_1-1)^2+4(x_2-3)^2$ subjected to $x_1^2+x_2^2= 5$

UNIT-V

6. a) i) What are the essential characteristics of dynamic programming. (10M)
ii) Set up the recursive relation, using dynamic programming approach, when an n-stage objectivefunction is to be maximized

OR

b) Solve the following LPP by dynamic programming approach. (10M)
Maximise $z = 2x_1 + 5x_2$
subject to$2x_1 + x_2 \leq 43$, $2x_2 \leq 46$, $x_1, x_2 \geq 0$.

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PART –A (10x 2Marks=20M)

1. a) What is variable in OT? (2M)
   b) Write the r^{th} differential of a multivariable function f(x). (2M)
   c) State an LP problem in standard form (2M)
   d) Explain the significance of Surplus variables with example. (2M)
   e) Write the mathematical model of transportation problem. (2M)
   f) How is degeneracy resolved in transportation problems. (2M)
   g) What is a one-dimensional minimization problem? (2M)
   h) why the study of unconstrained minimization methodsis important? (2M)
   i) What is the need for dynamic programming how is it different from linear programming? (2M)
   j) Define multistage decision method in DP. (2M)

PART –B (5x10Marks=50M)

UNIT-I

2. a) State the optimization problem. What are the various applications of optimization problems? Classify and explain various types of optimization problems with examples. (10M)

OR

   b) Use Kuhn-Tucker conditions to solve Max Z= $2x_1^2 + 12x_1x_2 - 7x_2^2$ subject to $2x_1+5x_2 <= 98$ and $x_1,x_2 >= 0$. (10M)

UNIT-II

3. a) i) How is the pivot reduction method applied for finding the solutions of linear simultaneous equations? (10M)
   ii) Discuss simplex algorithm wrt LPP

OR
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b) A progressive university has decided to keep its library open round the clock and gathered that the following number of attendants are required to reshelve the books: (10M)

Time of Day (hours)	Minimum Number of Attendants Required
0–4	4
4–8	7
8–12	8
12–16	9
16–20	14
20–24	3

If each attendant works eight consecutive hours per day, formulate the problem of finding the minimum number of attendants necessary to satisfy the requirements above as a LP problem

UNIT-III

4. a) i)State the assumptions in the transportation problem. (10M)
ii)What do you mean by an unbalanced transportation problem and explain how to convert the unbalanced transportation problem into a balanced one.

OR

b) Obtain initial solution in the following transportation problem by using VAM. (10M)

Source	D1	D2	D3	D4	D5	Availability
S1	5	3	8	6	6	1100
S2	4	5	7	6	7	900
S3	8	4	4	6	6	700
Requirement	800	400	500	400	600	

UNIT-IV

5. a) Using the one-dimensional search procedure, solve the following non-linear problem. Maximize f (x) =6x – x^2 by taking the initial upper and lower bounds at x = 4.8 and x = 0 and error tolerance = 0.04. (10M)

OR

b) i) What is the difference between the interior and extended interior penalty function methods? (10M)
ii) Discuss the procedure to check the convergence of constrained optimization problems.

UNIT-V

6. a) Explain the general algorithm for solving a dynamic programming problem. (10M)

OR

b) Minimize $z = y_1^2 + y_2^2 + y_3^2$ subjected to $y_1 + y_2 + y_3 = 10$ and $y_1, y_2, y_3 \geq 0$ using Bellman's principle. (10M)

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PART –A (10x 2Marks=20M)

1. a) Difference between a bound point and a free point in the design space (2M)
   b) Define a saddle point and indicate its significance (2M)
   c) Why is linear programming important in several types of industries? (2M)
   d) What is the difference between a slack and a surplus variable? (2M)
   e) How a transportation problembecomes a special case of LP problem? (2M)
   f) What is degeneracy in transportation problems? (2M)
   g) Differentiate between constrained and unconstrained problem. (2M)
   h) What is a descent method? (2M)
   i) What is a multistage decision problem? (2M)
   j) Define separable function. (2M)

PART –B (5x10Marks=50M)

UNIT-I

2. a) i)What are the methods for optimization with equality constraints. Discuss them with neat flow chart. (10M)
   ii)What are the differences between a constraint surface and a composite constraint surface?

OR

   b) Solve the following non-LPP by Lagrangian multiplier method: (10M)

$$Min\ Z = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$$
$$st\ x_1 + x_2 + x_3 = 15 \qquad \text{and } x_i \geq 0$$
$$2x_1 - x_2 + 2x_3 = 20$$
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UNIT-II

3\. a) i) Use simplex method to solve the following LP problem minimize $Z=5x+6y$ subject to the following constraints: $2x+5y\geq1500$; $3x+y\geq 1200$ and $x, y\geq0$. (10M)
ii) Can the Simplex method find the global solution to convex set programming problems?

OR

b) A firm manufactures 3 products A, B and C. The profits are Rs.3, 2 and 4, respectively. The firm has two machines and below is the processing time required in minutes on each machine foreach product. (10M)

		Product		
		A	*B*	*C*
Machine	*C*	4	3	6
	D	3	2	4

Machine C and D have 2,000 and 2,500 machine minutes, respectively. The firm must manufacture100 A's, 200 B's and 500 C's, but not more than 150 A's. Set up an linear programming problem to maximize profit

UNIT-III

4\. a) i)Describe the matrix form of the transportation problem. Illustrate with 2 origins and 3 destinations (10M)
ii)What do you understand by a balanced and an unbalanced transportation problem? How an unbalanced problem is tackled?

OR

b) Determine an initial basic feasible solution to the following transportation problem using north- west corner rule. (10M)

	To					*Available*
From	3	4	6	8	9	20
	2	10	1	5	8	30
	7	11	20	40	3	15
	2	1	9	14	16	13
Demand	40	6	8	18	6	

UNIT-IV

5\. a) By using one-dimensional search procedure, maximize $f(x) = 20x - 3x^2 - x^4$ taking error tolerance as 0.01 (10M)

OR

b) Classify the constrained optimization techniques and briefly explain each technique. (10M)

UNIT-V

6. a) i)Write short note on characteristics of dynamic programming. (10M)
ii) State two engineering examples of serial systems that can be solved by dynamic programming.

OR

b) Use Bell man's principle of optimality to find the optimum solution to the following problem (10M)

$$\text{Minimize } z = y_1^2 + y_2^2 + y_3^2$$

$$\text{S.T } y_1 + y_2 + y_3 \geq 15,$$

$$y_1, y_2, y_3 \geq 0$$

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PART –A (10x 2Marks=20M)

1.	a)	Define Trajectory optimization problem	(2M)
	b)	What is an active constraint?	(2M)
	c)	State four applications of linear programming	(2M)
	d)	Define the in feasibility form	(2M)
	e)	Write the mathematical model of transportation problem.	(2M)
	f)	Can degeneracy occur in a transportation problem? Justify.	(2M)
	g)	What is Fibonacci method?	(2M)
	h)	State the iterative approach used in unconstrained optimization.	(2M)
	i)	What is the need for dynamic programming how is it different from linear programming?	(2M)
	j)	Why are the components numbered in reverse order in dynamic programming?	(2M)

PART –B (5x10Marks=50M)

UNIT-I

2.	a)	i) What are the various applications of optimization problems? ii) What is the significance of the conditions of variables in optimization problems?	(10M)
		OR	
	b)	i)Find the maxima and minima, if any, for functionf (x) = $4x^3 – 18x^2 +27x-7$ ii)State and explain the necessary conditions for existence of relative optima in case of multivariable objective functions with and out constraints.	(10M)

UNIT-II

3.	a)	i) How is the pivot reduction method applied for finding the solutions of linear simultaneous equations? ii) Can the Simplex method find the global solution to convex set programming problems? Explain.	(10M)
		OR	
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b) A paper mill received an order for the supply of paper rolls of widths and lengths as indicated below. (10M)

Number of Rolls Ordered	Width of Roll (m)	Length (m)
1	6	100
1	8	300
1	9	200

The mill produces rolls only in two standard widths, 10 and 20 m. Themill cuts the standard rolls to size to meet the specifications of theorders. Assuming that there is no limit on the lengths of the standardrolls, find the cutting pattern that minimizes the trim losses while satisfyingthe order above.

UNIT-III

4. a) i) Give the three variations in transportation problem. (10M)
ii)Describe a general transportation problem. Explain how to determine an initial basic feasible solution to the problem using Vogel's method.

OR

b) Determine the optimal solution for the transportation matrix given below.Is the solution unique? If not give alternate solution. (10M)

Warehouses

Plants	W	X	Y	Z	Supply
P	11	12	13	14	10
Q	21	22	23	24	6
R	31	32	33	34	15
Rquirements	2	9	3	17	

UNIT-IV

5. a) Find by Fibonacci method Minimum of $f(x) = x^3-3x+5$, $0\leq x \leq 1.2$ within aninterval of uncertainty 0.35 L_0 where L_0 is the original interval ofuncertainty. (10M)

OR

b) Classify the constrained optimization techniques and briefly explain each technique. (10M)

UNIT-V

6. a) i)Explain the concept of dynamic programming. (10M)
ii)Explain the following in the context of dynamic programming
(i) Principle of optimality (ii) State (iii) Stage

OR

b) Use dynamic programming technique to solve the following problem. (10M)
Max $Z = X1.X2.X3.X4$
Subject to $X1 + X2 + X3 + X4 = 12$
$X1, X2, X3, X4 \geq 0$